

0602221-AI Programming

Lab Exercise 03

Main Topics

- Boolean expressions
- If statements
- If-else statements
- While loops

Exercise

This lab is designed to give you more practice using if-else statements, nested if statements, and an introduction to the while loop.

Getting Started

To start this exercise, you should:

1. Open your Python development environment, such as IDLE, VS Code, PyCharm, or Jupyter Notebook. Lab03.
2. Create a new Python file named **lab03.py**.

Problem Description

For this lab you are to write a complete Python program to run a simple cash register for an eye glass store. A customer can buy either a pair of single prescription glasses (at a cost of \$40.00), or a pair of non-prescription sunglasses (at a cost of \$25.00). On each pair of single prescription glasses the customer must choose either an anti-glare coating (at a cost of \$12.50) or a brown tint coating (at a cost of \$9.99).

Requirements

At each point where the user is presented with an option:

- Display a simple menu with options.
- Repeatedly prompt the user until a **valid** option is entered.
- Use the valid input to guide the program logic.

Sample Run

What kind of glasses would you like:

1 -> prescription, 2 -> non-prescription : 0

1 -> prescription, 2 -> non-prescription : 3

1 -> prescription, 2 -> non-prescription : 1

What kind of coating would you like:

1 -> anti-glare, 2 -> brown tint : 0

1 -> anti-glare, 2 -> brown tint : 3

1 -> anti-glare, 2 -> brown tint : 1

Your total cost is \$52.5

Submission Instructions

Once you have written your program:

1. Ensure your program compiles and runs without errors or warnings.
2. Test your program multiple times to verify correctness for all choices.
3. If correct, present your work to me for check-off.